

- 1 Phosphatases are enzymes that catalyse the removal of phosphate from other molecules, releasing inorganic phosphate (P_i). A phosphatase can be extracted from mung bean seedlings, *Vigna radiata*.

Fig. 1.1 shows mung bean seedlings.

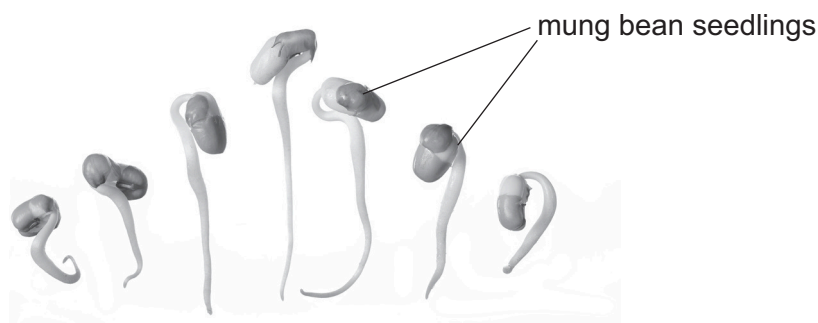


Fig. 1.1

A student found a published method to extract phosphatase from mung bean seedlings.

- Use a pestle and mortar to grind mung bean seedlings, with a small volume of distilled water, to make a paste.
- Add distilled water to the paste to make a mixture with a total volume of 50 cm^3 .
- Filter the mixture and put the filtrate into a clean centrifuge tube.
- Centrifuge the filtrate until a solid pellet is formed, as shown in Fig. 1.2.
- Pour the liquid extract containing phosphatase into a clean test-tube.

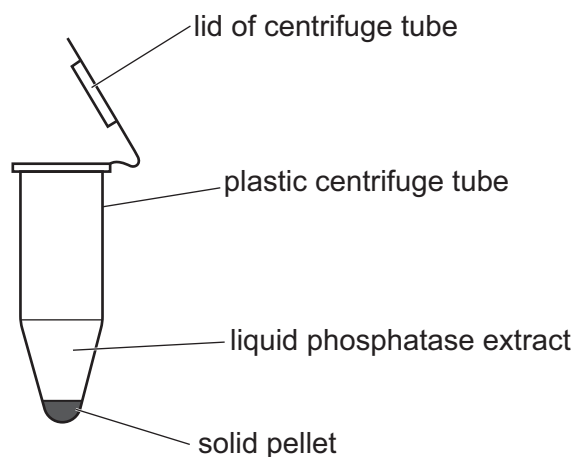


Fig. 1.2

- (a) State **two** other variables that should be standardised in the published method so that extracts with the same concentration of phosphatase can be produced.

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- (b) The student read that the optimum pH for phosphatase extracted from mung bean seedlings is less than pH7.0.

The student decided to investigate phosphatase activity, using the substrate phenolphthalein phosphate (PPP).

In the first four steps of the method, the student:

- mixed 5 cm³ of pH6.0 buffer solution with 1 cm³ of a 1.0% solution of PPP in a test-tube
- added 1 cm³ of the phosphatase extract to the test-tube and started a timer
- incubated the test-tube for 10 minutes in a water-bath at 30 °C
- stopped the enzyme reaction after 10 minutes, by adding 5 cm³ of 10.0% solution of sodium carbonate. Sodium carbonate solution is alkaline.

- (i) The buffered PPP solution and the phosphatase extract were mixed and then placed in the water-bath. The student identified this as a source of error in the method.

Explain why this is a source of error **and** state how you would modify the method to remove this source of error.

explanation

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modification

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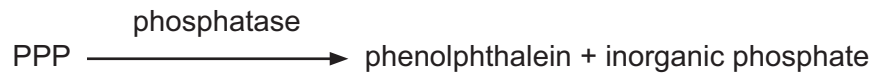
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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (ii) Suggest how the addition of 10.0% solution of sodium carbonate stops the enzyme reaction.

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- (c) The phosphatase catalyses the removal of inorganic phosphate from PPP as shown by:



The 10.0% solution of sodium carbonate added at the end of the experiment also causes any phenolphthalein to turn pink. The intensity of the pink colour is an indication of the concentration of phenolphthalein.

To estimate the concentration of phenolphthalein produced by the reaction, the student decided to make a proportional dilution using a 2.0% stock solution of phenolphthalein.

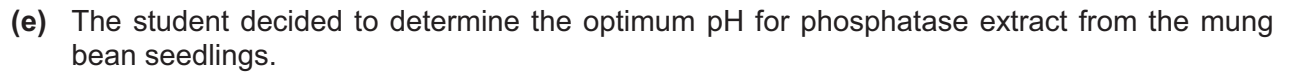
The student made 50 cm³ of each diluted solution.

Describe a method the student could use to make a proportional dilution of the 2.0% stock solution of phenolphthalein to get a range of concentrations.

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- (d) Describe how the student could use the dilutions from (c) and a colorimeter to estimate the concentration of phenolphthalein in a reaction mixture.

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Your method should be set out in a logical order and be detailed enough to allow another person to follow it.

[6]

(f) The student investigated the effect of a competitive inhibitor on the activity of a phosphatase.

Fig. 1.3 shows a graph of the initial rates of enzyme reaction against PPP concentration for the enzyme with no inhibitor and the enzyme with inhibitor.

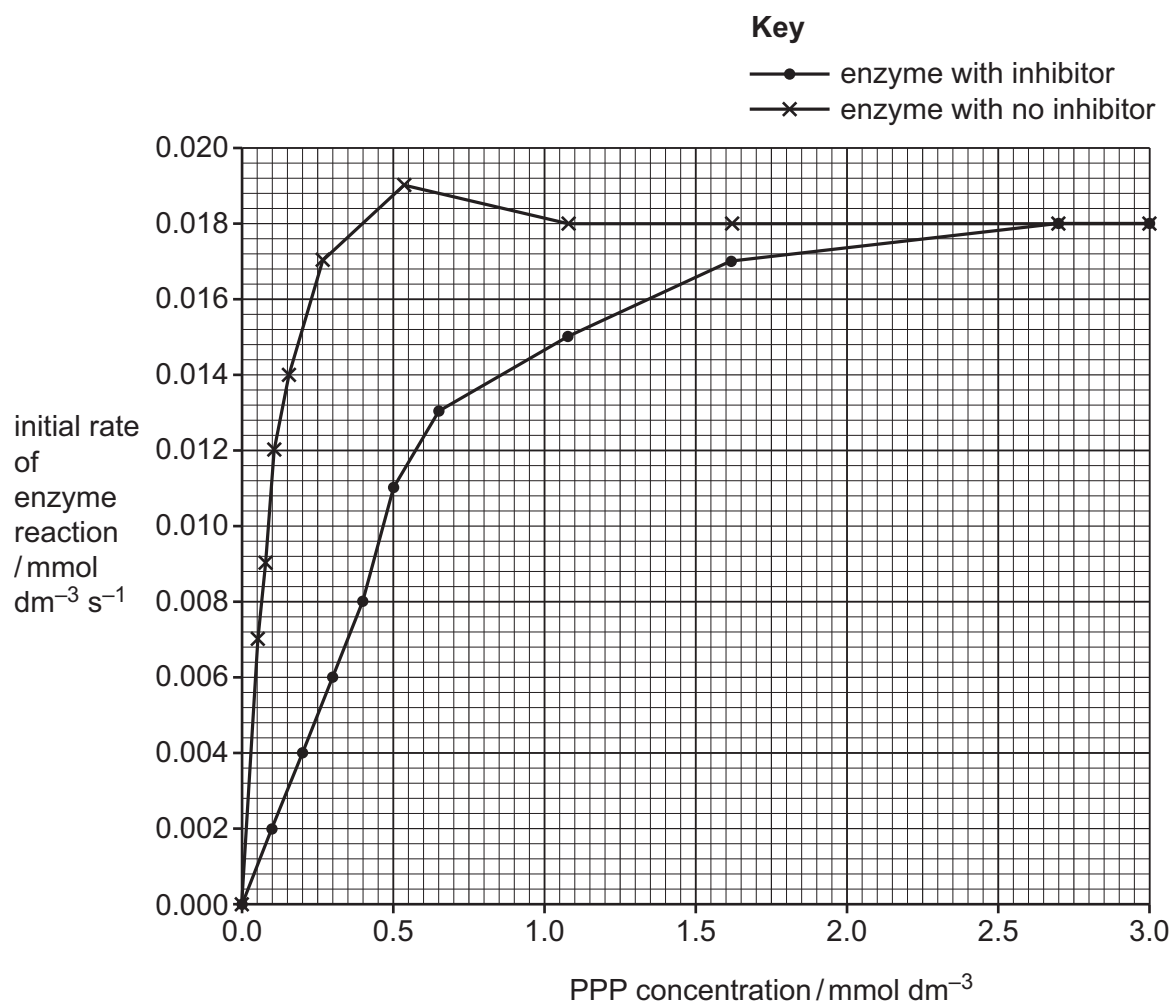


Fig. 1.3

(i) One of the data plots in Fig. 1.3 is anomalous.

Circle the anomalous data plot in Fig. 1.3 **and** explain why you think it is anomalous.

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (ii) Use Fig. 1.3 to determine the Michaelis–Menten constant (K_m) for the enzyme with no inhibitor and for the enzyme with inhibitor.

Include the correct units in your answers.

K_m for the enzyme with no inhibitor =

K_m for the enzyme with inhibitor = [1]

- (iii) Calculate the percentage increase in the K_m that occurs in the presence of the inhibitor.

Show your working.

percentage increase = % [2]

[Total: 20]